

6.2 Data Warehouse and Data Mining

A. SYLLABUS

Course No.	Title of the Course	Credits	Course Structure	Pre-Requisite
CA CSC 6 02	Data Warehouse and Data Mining	4	3L-0T-2P	DBMS
COURSE OUTCOMES: CO 1 : To understand the basic principles, concepts and applications of data warehouse and data mining. CO 2 : To design task of data mining as an important phase of the knowledge recovery process. CO 3 : To analyse Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment and design a data warehouse or data mart to present information needed by management in a form that is usable for management clients. CO 4 : To apply concepts that provide the foundation of data mining CO 5 : To understand the concepts of Data Mining algorithms for classification applications				

Unit No.	Topics
Unit 1	Introduction to Data Warehousing: Overview, Difference between Database System and Data Warehouse, The Compelling Need for data warehousing, Data warehouse – The building Blocks: Defining Features, data warehouses and data marts, overview of the components, Three tier architecture, Metadata in the data warehouse. Data pre-processing: Data cleaning, Data gateway, Synchronization of databases, Data transformation and ETL Process, ETL tools, interoperability of data and applications.
Unit 2	Dimensional Modelling: Defining the business requirements: Dimensional analysis, information packages – a new concept, requirements gathering methods, requirements definition: scope and content. Principles of Dimensional Modelling: Objectives, From Requirements to data design, Multidimensional Data Model, Schemas: the STAR schema, the Snowflake schema, fact constellation schema.
Unit 3	OLAP in the Data Warehouse: Demand for Online Analytical Processing, limitations of other analysis methods- OLAP is the answer, OLAP definitions and rules, OLAP characteristics, major features and functions, hyper cubes. OLAP Operations: Drill-down and roll-up, slice-and-dice , pivot or rotation, OLAP models, overview of variations, the MOLAP model, the ROLAP model, the DOLAP model, ROLAP versus MOLAP, OLAP implementation considerations. Query and Reporting, Executive Information Systems (EIS), Data Warehouse and Business Strategy. Advanced Data warehouse Techniques: Document oriented NoSQL Databases- MangoDB, CouchDB; Graph Databases-Neo4J, Infinite Graph.
Unit 4	Data Mining Basics: What is Data Mining, Data Mining Defined, The knowledge discovery process (KDD Process), Data Mining Applications- The Business Context of

	Data Mining, Data Mining for Process Improvement, Data Mining as a Research Tool, Data Mining for Marketing, Benefits of data mining, Major Data Mining Techniques: Classification and Prediction: Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, KNN Algorithm.
Unit 5	Data Mining Algorithms: Cluster detection, K- means Algorithm, Outlier Analysis, memory-based reasoning, link analysis, Mining Association Rules in Large Databases: Association Rule Mining, genetic algorithms, neural networks, Data mining tools.
Suggested Readings:	
1. Paul Raj Poonia, “Fundamentals of Data Warehousing”, John Wiley & Sons. 2. Kamber and Han, “Data Mining Concepts and Techniques”, Hart Court India P. Ltd. Elsevier Publications Second Edition. 3. W. H. Inmon, “Building the operational data store”, 2nd Ed., John Wiley. 4. Shmueli, “Data Mining for Business Intelligence : Concepts, Techniques and Applications in Microsoft Excel with XLMiner”, Wiley Publications.	

CO-PO & CO-PSO MAPPING TABLE

CO PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	1	2	1	1	0	1	0	0	0	1	1		2	1
CO2	2	1	1	2	3	2	0	0	0	2	1	1	2	2	1
CO3	3	1	1	2	1	0	1	1	1	1	3	2	1	2	1
CO4	1	1	2	1	1	1	0	0	1	1	2	1	1	2	1
CO5	2	2	2	2	0	1	0	0	1	1	1	1	1	2	1

C. THEORY LECTURE PLAN

S.N o.	CONTENT	NUMBER OF LECTURE S	Unit
1	Introduction to Data Warehousing: Overview, Difference between Database System and Data Warehouse, The Compelling Need for data warehousing	1	UnitI (6)
2	Data warehouse – The building Blocks: Defining Features, data warehouses and data marts, overview of the components, Three-tier architecture, Metadata in the data warehouse.	1	
3	Data pre-processing: Data cleaning, Data gateway, Synchronization of databases, Data transformation	2	

4	ETL Process, ETL tools, interoperability of data and applications	2	
CLASS TEST-I			
5	Dimensional Modelling: Defining the business requirements: Dimensional analysis	2	UnitII (9)
6	Information packages – a new concept, requirements gathering methods, requirements definition: scope and content.	2	
7	Principles of Dimensional Modelling: Objectives, From Requirements to data design, Multidimensional Data Model	2	
8	STAR schema	1	
9	Snowflake schema	1	
10	Fact constellation schema	1	
MID SEMESTER EXAMS			
11	Demand for Online Analytical Processing, limitations of other analysis methods	1	UnitIII (12)
12	OLAP definitions and rules, OLAP characteristics, major features and functions, hyper cubes	2	
13	OLAP Operations: Drill-down and roll-up, slice-and-dice , pivot or rotation	1	
14	OLAP models, overview of variations, the MOLAP model, the ROLAP model, the DOLAP model, ROLAP versus MOLAP	2	
15	OLAP implementation considerations	1	
16	Query and Reporting	1	
17	Executive Information Systems (EIS), Data Warehouse and Business Strategy	1	
18	Advanced Data warehouse Techniques	1	
19	MangoDB, CouchDB	1	
20	Graph Databases-Neo4J, Infinite Graph	1	
CLASS TEST-II			
21	Data Mining Basics: What is Data Mining, Data Mining Defined, The knowledge discovery process (KDD Process)	2	UnitI V (8)
22	The Business Context of Data Mining, Data Mining for Process Improvement	1	
23	Data Mining as a Research Tool, Data Mining for Marketing. Benefits of data mining	1	

24	Classification and Prediction: Issues Regarding Classification and Prediction	1	UnitV (6)
25	Classification by Decision Tree Induction	2	
26	KNN Algorithm	1	
27	Cluster detection, K- means Algorithm	1	
28	Outlier Analysis, memory-based reasoning, link analysis, Data Mining Tools	1	
29	Mining Association Rules in Large Databases: Association Rule Mining	1	
30	Genetic algorithms	1	
31	neural networks	1	

D. PRACTICAL CLASS PLAN

Lab Class No.	Name of the Experiment
1	Create an Employee Table with the help of Data Mining Tool WEKA.
2	Apply pre-processing to the training data-set of Employee table.
3	Normalize Employee Table data using knowledge flow.
4	Finding Association rules for banking data/Employee data.
5	To construct decision tree for employee/weather data and classify it.
6	Write a procedure for visualization for weather table.
7	Write a procedure for cross-validation using J48 algorithm for weather table.
	MID SEMESTER EVALUATION
8	Write a procedure for clustering buying data using Cobweb Algorithm.
9	Write a procedure for clustering weather data using EM Algorithm.
10	Write a procedure for clustering customer data using simple K-means algorithm.
	END SEMESTER EVALUATION

E. SELF STUDY

Sr. No.	Topic	
1.	Write a procedure for employee data using make density based cluster algorithm.	Unit IV
2.	Write a procedure for banking data using Farthest first algorithm.	Unit V